1

Islamic University of Technology

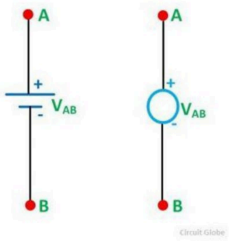
Course no: Phy 4143
Course Title: Physics II
Course Teacher : Dr Md. Sahab Uddin
Professor

2

Voltage source, current source and transformation

A Source is a device which converts mechanical, chemical, thermal or some other form of energy into electrical energy. In other words, the source is an active network element meant for generating electrical energy.

- Voltage Source
- Current Source



Voltage source: A voltage source is a two-terminal device whose voltage at any instant of time is constant and is independent of the current drawn from it. Such a voltage source is called an Ideal Voltage Source and have zero internal resistance.

The example of voltage sources is batteries and alternators.

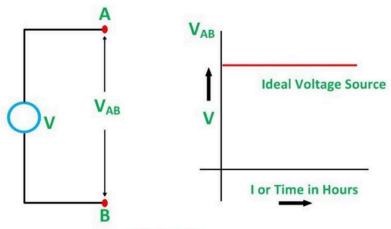


Figure A

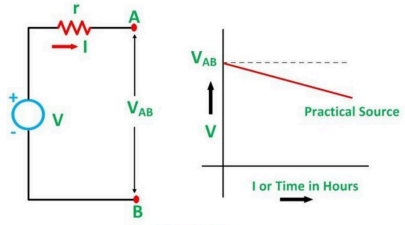
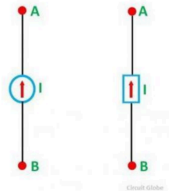


Figure B

3

An ideal current source is a two-terminal circuit element which supplies the same current to any load resistance connected across its terminals. It is important to keep in mind that the current supplied by the current source is independent of the voltage of source terminals. It has infinite resistance.



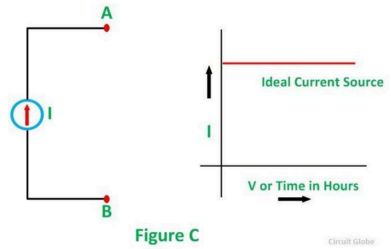


Figure C

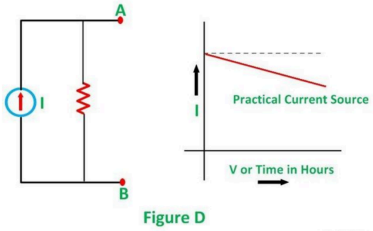
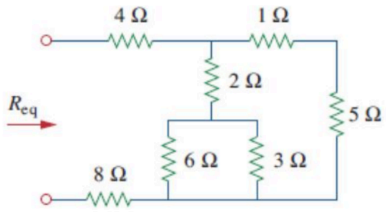


Figure D

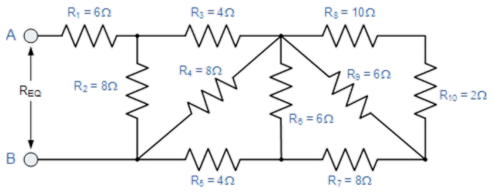
The example of current sources is photoelectric cells, collector currents of transistors.

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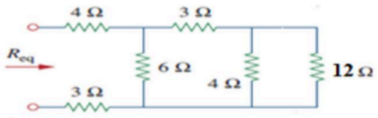
nd the Req for the circuit shown in below figure.



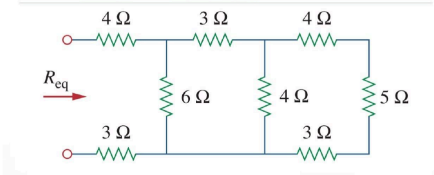
2) Find the Req for the circuit shown in below figure.



3) Find the Req for the circuit shown in below figure.



4) Find the Req for the circuit shown in below figure.

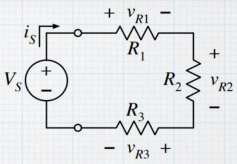


8

Voltage Divider Rule

Voltage divider

Consider a portion of circuit that has several resistors in series, like the circuit at right. Suppose we want to find the voltage across R_2.



We could start by finding the current, which would be equal to the source voltage divided by the equivalent resistance of the string.

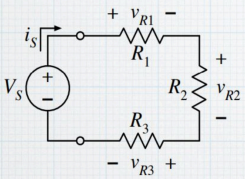
$$i_S = \frac{V_S}{R_{eq}}$$

For the series string, $R_{eq} = R_1 + R_2 + R_3$.

Then the voltage across R_2 is just

$$v_{R2} = R_2 \cdot i_S = \frac{R_2}{R_{eq}} \cdot V_S = \frac{R_2}{R_1 + R_2 + R_3} \cdot V_S$$

The other resistor voltages are calculated just as easily.



$$v_{R2} = \frac{R_2}{R_1 + R_2 + R_3} \cdot V_S$$

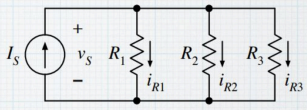
$$v_{R3} = \frac{R_3}{R_1 + R_2 + R_3} \cdot V_S$$

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Current Divider Rule

Current divider

Same idea, but with parallel resistors dividing a current. Suppose we want to know the current through R_2.



We could start by finding the voltage, which would be equal to the source current multiplied by the equivalent resistance of the parallel resistors.

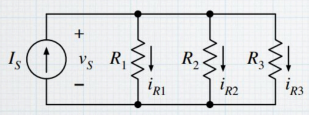
$$v_S = I_S \cdot R_{eq}$$

For the parallel combination, $R_{eq} = \dfrac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$

Then the current through R_2 is

$$i_{R2} = \frac{v_S}{R_2} = \frac{R_{eq}}{R_2} \cdot I_S = \frac{\frac{1}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \cdot I_S$$

The other resistor currents are calculated just as easily.



$$i_{R1} = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \cdot I_S$$

$$i_{R3} = \frac{\frac{1}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \cdot I_S$$